Superconducting nanowires at the ePIC far-forward: what a closer approach would buy the polarized-lithium tags

2026-08-26 · companion to plans/09 and to open questions #19 and #20 · computed with `evgen/scripts/nearbeam_aperture_scan.py`, `nearbeam_reach_gain.py` and `nearbeam_sensor_budget.py` on `polligen.reco`, `polligen.recopseudo` and `polli_fastsim.spectator` · primary sources in `refs/`

The far-forward tags of the polarized-lithium programme are beam-blind. An intact ^{6}Li has A/Z = 2 exactly and the α from its breakup has 0.99813 of the beam's rigidity, so neither is separated from the beam by dispersion: what separates them is the transverse *angle* at the interaction point. The acceptance is therefore exp(−B|t|) evaluated at |t| = (θp)² — an exponential in the *square* of the approach angle times the square of the momentum — and approach distance is the whole measurement. Superconducting nanowire detectors, developed for nuclear physics by the Argonne MEP and Physics Divisions, are the candidate technology. This note prices what a closer approach is worth, asks whether a nanowire can deliver it, and separates the part of the case that survives scrutiny from the part that does not.

> **Three findings.** (1) **The aperture argument is not where the value is.** A closer approach is worth a great deal — ×26 in coherent acceptance at 5 × 41, ×569 at 10 × 100, and the difference between a fit that returns a_t = 0.48 ± 1.45 and one that returns four clean |t| bins — but the 10σ rule that sets the limit is *machine protection and halo*, not sensor survival, so a more radiation-hard sensor does not relax it. Two things make the aperture case weaker still: the ePIC pot geometry has changed since the snapshot this repository measured (§6), and a cryostat makes the package bigger, not smaller (§5). (2) **Charge identification is where the value is.** A current-biased nanowire is not a proportional device — it latches, and its pulse amplitude is the same for every species — but its *firing threshold* is set by the deposit through $I_{th}/I_c = 1 - 2r_s/w$ with $r_s \propto z$. At the 1 μm wire width that already exists, ^{6}Li, α and d have three separated turn-on currents, and two bias points tag Z by the firing pattern alone (§4). This is the observable the AC-LGAD chain cannot supply and nothing else at IP6 answers — open question #19. (3) **The blockers are thermodynamic and areal, not physics.** A ~1 W cold stage against an estimated 8–16 W of beam-induced RF power; no precedent anywhere for a biased superconducting detector inside an accelerator's beam vacuum; radiation hardness unmeasured after four years of a funded programme; and a state of the art of 2 × 2 mm² against the tens of mm² a near-beam strip needs (§5).

1 · Why the angle is everything

Both far-forward lithium tags are rigidity-degenerate with the beam. The coherent measurement needs the intact ^{6}Li, whose A/Z is exactly the beam's; the tagged measurements need the α spectator, whose rigidity is 0.99813 of it. Neither is swept aside by the dipoles. The only handle is the transverse angle, and because the coherent recoil follows exp(−B|t|) with B ≈ 50 GeV^{-2} and $|t| = (\theta_x p)^2$, the tagged fraction falls as the exponential of the *square* of the approach angle times the *square* of the beam momentum. Halving the approach angle at 10 × 100 is not worth a factor of two; it is worth a factor of 569.

Two angles compete to set that approach. The **10σ envelope** — 0.727 mrad for the high-acceptance optics — is an operational retraction, a machine-protection rule about beam halo. The **geometric aperture** of the sensor package is what the silicon, its RF shield, its ASIC and its module granularity actually leave open. Shooting an intact ^{6}Li through the ePIC geometry (`tools/fullsim`) measured the second: $|\theta_x| \gtrsim$ 2.00 / 1.35 / 1.03 mrad at 5 × 41 / 10 × 100 / 18 × 275, or 27.5 / 18.6 / 14.2 σ_θ — a package that stops 4.2σ to 17.5σ short of where the machine would allow it. §6 revisits whether that measurement still describes today's geometry; the curves below do not depend on the answer.

2 · What a closer approach is worth

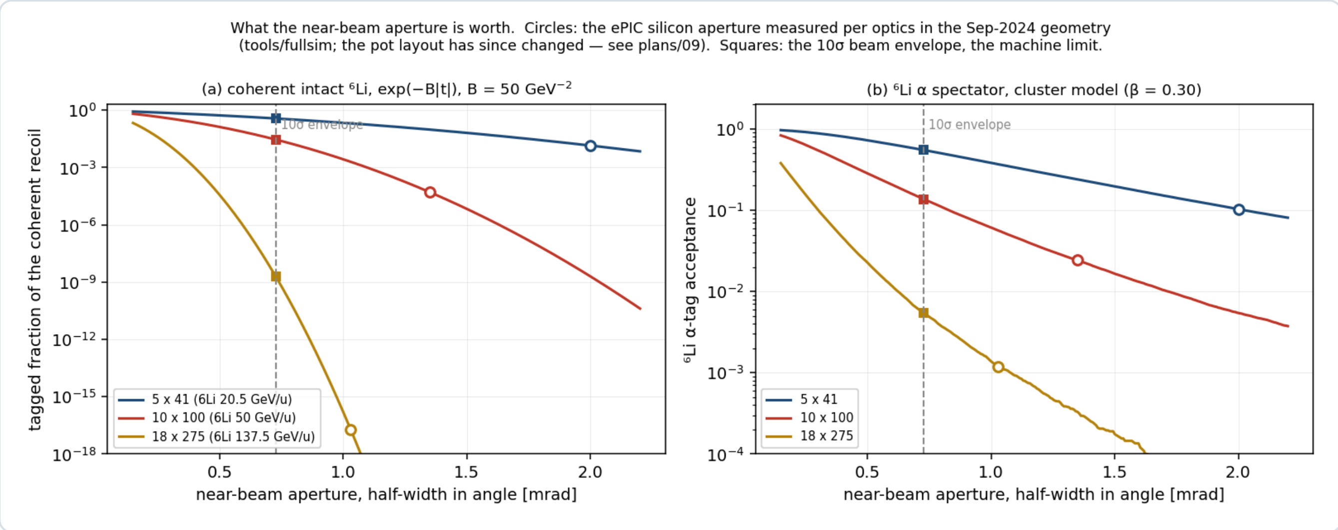

Figure 1 · The price of every aperture. Tagged fraction of the coherent intact ⁶Li (a) and of the ⁶Li α spectator (b) against the near-beam half-width in angle, per beam configuration. Circles mark the ePIC silicon aperture measured in the September-2024 geometry; squares mark the 10σ envelope. The vertical half-width is held at its measured value per optics (3.0 / 3.0 / 2.3 mrad). Note the vertical scale of (a): eighteen decades. `nearbeam_aperture_scan.py`.

The analytic gains are large and very unequal:

	5 × 41 (20.5 GEV/U)	10 × 100 (50)	18 × 275 (137.5)
coherent tagged fraction, silicon aperture	1.41×10^{-2}	5.12×10^{-5}	1.94×10^{-17}
coherent tagged fraction, 10σ envelope	3.71×10^{-1}	2.91×10^{-2}	1.97×10^{-9}
gain	**×26**	**×569**	$\times1.0\times10^{8}$, still dead
⁶Li α tag, silicon aperture	0.103	0.024	0.0012
⁶Li α tag, 10σ envelope	0.550	0.137	0.0054

Coherent: `reco.rp_hole_acceptance` at B = 50 GeV⁻² with the measured rectangular cutout. α tag: the cluster model of `polli_fastsim.spectator` at β = 0.30.

The top energy is not rescued, by any technology. At 825 GeV even a 10σ approach demands $p_T > 0.60$ GeV, $|t| > 0.36$ GeV², and $\exp(-50 \times 0.36) = 10^{-9}$. No aperture makes the coherent channel live at 18 × 275; the client there is the α tag and nothing else.

2.1 · Through the chain, not just the acceptance

Acceptance is not the measurement. Running the full coherent chain — Roman-Pot emulation, the two azimuths $\alpha = \varphi_e - \varphi_S$ and $\beta = \varphi_t - \varphi_S$, the spin-state-sorted 2-D harmonic fit — at both apertures shows three effects, not one.

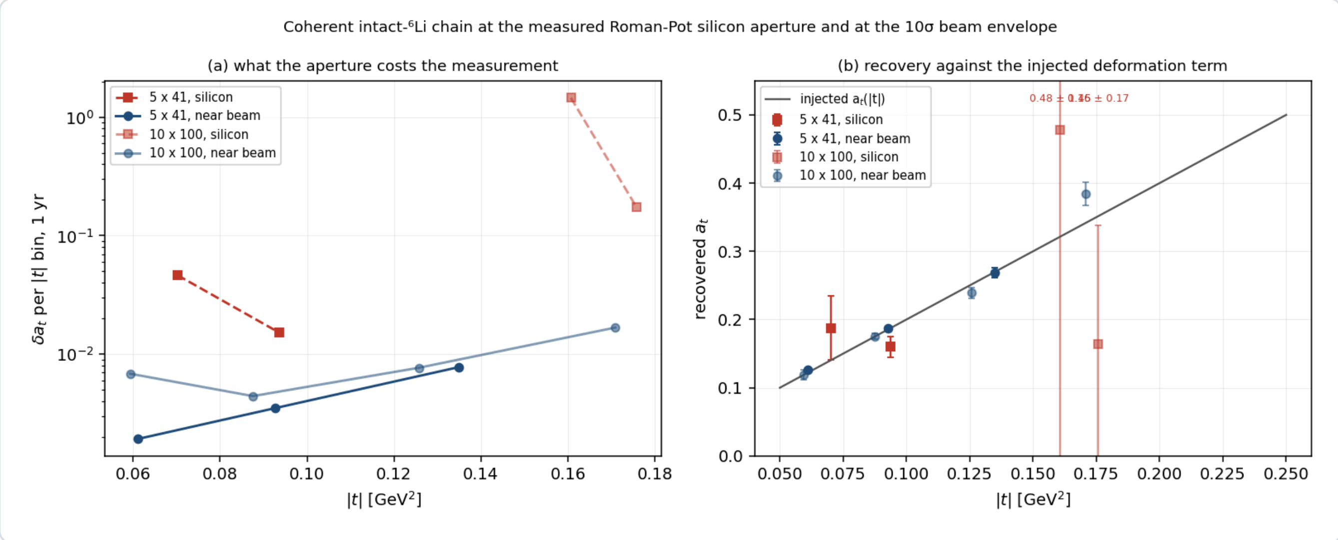

Figure 2 · What the aperture costs the measurement. (a) δa_t per |t| bin at 1 year, at the measured silicon aperture (dashed, squares) and at the 10σ envelope (solid, circles), for the two configurations where the coherent channel is alive. (b) The recovered a_t against the injected deformation term; the off-scale silicon point at 10 × 100 is labelled. `nearbeam_reach_gain.py`.

OPTICS	APERTURE	ACCEPTANCE	|T| BINS	ΔA_T PER BIN, 1 YR	A_E (INJECTED 0.0100)
5 × 41	2.00 mrad, silicon	1.43×10^{-2}	2 of 4	0.046, 0.015	−0.0060 ± 0.0045, +0.0197 ± 0.0044
5 × 41	0.727 mrad, 10σ	3.62×10^{-1}	3 of 4	0.0019, 0.0035, 0.0078	+0.0084 ± 0.0016, +0.0087 ± 0.0031, +0.0012 ± 0.0076
10 × 100	1.35 mrad, silicon	1.02×10^{-4}	2 of 4	1.45, 0.175	−0.068 ± 0.049, −0.019 ± 0.020
10 × 100	0.727 mrad, 10σ	3.25×10^{-2}	4 of 4	0.0069, 0.0044, 0.0077, 0.0168	+0.0117 ± 0.0013 … +0.0304 ± 0.0118

1. **Statistics.** δa_t improves ×8 to ×40 where the measurement already exists.
2. **A configuration.** At 10 × 100 the silicon aperture returns $a_t = 0.48 \pm 1.45$ against a truth of 0.32 — not a measurement. At 10σ it returns four clean bins. The mid-energy configuration goes from dead to the best-covered in the programme, which matters because the reconstruction-chain report otherwise concludes that the coherent channel is a low-energy-only programme. *That conclusion is aperture-conditional, not physics-conditional.*
3. **Bias, not only error.** At 5 × 41 with the silicon aperture the gluon-transversity coefficient comes back −0.0060 ± 0.0045 and +0.0197 ± 0.0044 against an injected 0.0100 — both wrong by more than 2σ, because the emptied azimuthal circle leaves the seven harmonic columns nearly degenerate. At 10σ every bin is consistent. A tight aperture does not merely cost events; it biases the extraction.

> **A trap this exposed.** Money plot 6R carries `cut_scale_x = 2.5` from the pre-measurement belief that the pots surround a *wide horizontal slot*. With the geometric aperture measured, carrying 2.5 as well imposes a 25σ horizontal retraction that no machine requirement asks for — and it binds *before* the geometry does, hiding the entire effect. Both new scripts hold the envelope at 10σ in each axis and let the measured geometry be the only aperture.

3 · The idea is already on record

Near-beam recoil tagging is not a new application to propose to the Argonne group. It is theirs, twice over. The EIC Yellow Report §14.5.3 lists the four SNSPD applications the EIC R&D committee identified, of which the first is *"a Roman pot detector in the forward region about 35 meters or more from the interaction point to tag low momentum transfer recoiling ions"* [1]. Armstrong's FY22 EIC generic-detector-R&D proposal — reviewed and funded — states the mechanism verbatim: *"The so-called 10 σ rule-of-thumb for Roman pots prevents the beam from damaging the detectors by limiting its proximity to the beam. Radiation hard nanowire detectors will be able to move closer to the beam, thus extending the acceptance reach at lower t"* [2].

What this study adds is the number attached to *lower t*, for a species whose acceptance is exponentially more sensitive to it than the DVCS protons that motivate the rest of the far-forward programme. Table 2 is that number.

> **But the stated mechanism does not hold.** The 10σ rule is a machine-protection and beam-halo rule, not a statement about sensor damage: it exists to keep the *beam* safe and to keep the detector out of the halo, and a more radiation-hard sensor does not relax it. The ePIC geometry says as much in the file itself — the per-energy insertions are *"the ten-sigma cuts for the Roman pots, translated to the physical layout we currently have"* [3]. The honest version of the argument is **granularity and dead edge**, not radiation hardness: the insertion is quantised by the module, and the innermost step is held back to protect a block one module wide. A millimetre-scale tile whose active area reaches within a few hundred nanometres of the substrate edge [1] — measured, in the SMSPD case, to be sharper than the 10 μm telescope used to look at it [8] — is exactly the object that de-quantises that staircase. That is a pot-mechanics argument, and it is worth making; it is not the argument the proposal makes.

4 · Charge identification: where the technology does something nothing else does

Open question #19 has no answer at IP6. An intact ^{6}Li, an α and a d from breakup share rigidity *and* velocity; only dE/dx separates them, going as z^2 — 9 : 4 : 1 — and the EICROC AC-LGAD chain records pulse amplitude for charge-sharing but no EIC document addresses Z-ID in the pots. The only documented concept is a z^2 Cherenkov behind the IR-8 secondary-focus pots.

A nanowire will not do it by pulse height. The device latches: the pulse amplitude is the diverted *bias* current, and both Fermilab/JPL beam papers show the amplitude distributions from 120 GeV hadrons, 120 GeV muons, 8 GeV pions and 8 GeV showering electrons to be indistinguishable, with one amplitude threshold serving all [7,8]. Any scheme that reads Z off a pulse height is dead on arrival.

It will do it by firing threshold. In the normal-core hot-spot regime a deposit Q makes a normal core of radius $r_s \propto \sqrt{Q}$, and the wire switches when that core spans it against the bias-suppressed superconducting width:

$$r_s = \sqrt{\frac{Q}{e\pi c\rho(T_c - T_0)}}\,, \qquad \frac{I_{\rm th}}{I_c} = 1 - \frac{2r_s}{w}$$

Three groups converge on this relation from three directions — Argonne's own Eqs. 1–2 for 120 GeV protons [4], Renema's $E = (w/C)^2(1 - I_b/I_c)^2$ for photons, and the ion literature's $I_{th}/I_c = 1 - (zeV)^{1/2}C/w$ [9]. Because dE/dx ∝ z^2 at fixed velocity, and because a ^{6}Li at 137.5 GeV/u has βγ = 148 against 128 for Argonne's calibration proton — the same velocity to 15% — **r_s scales as z linearly, with no velocity correction**: 134 nm (p, d), 268 nm (α), 402 nm (^{6}Li), anchored on Argonne's own measured 134 nm.

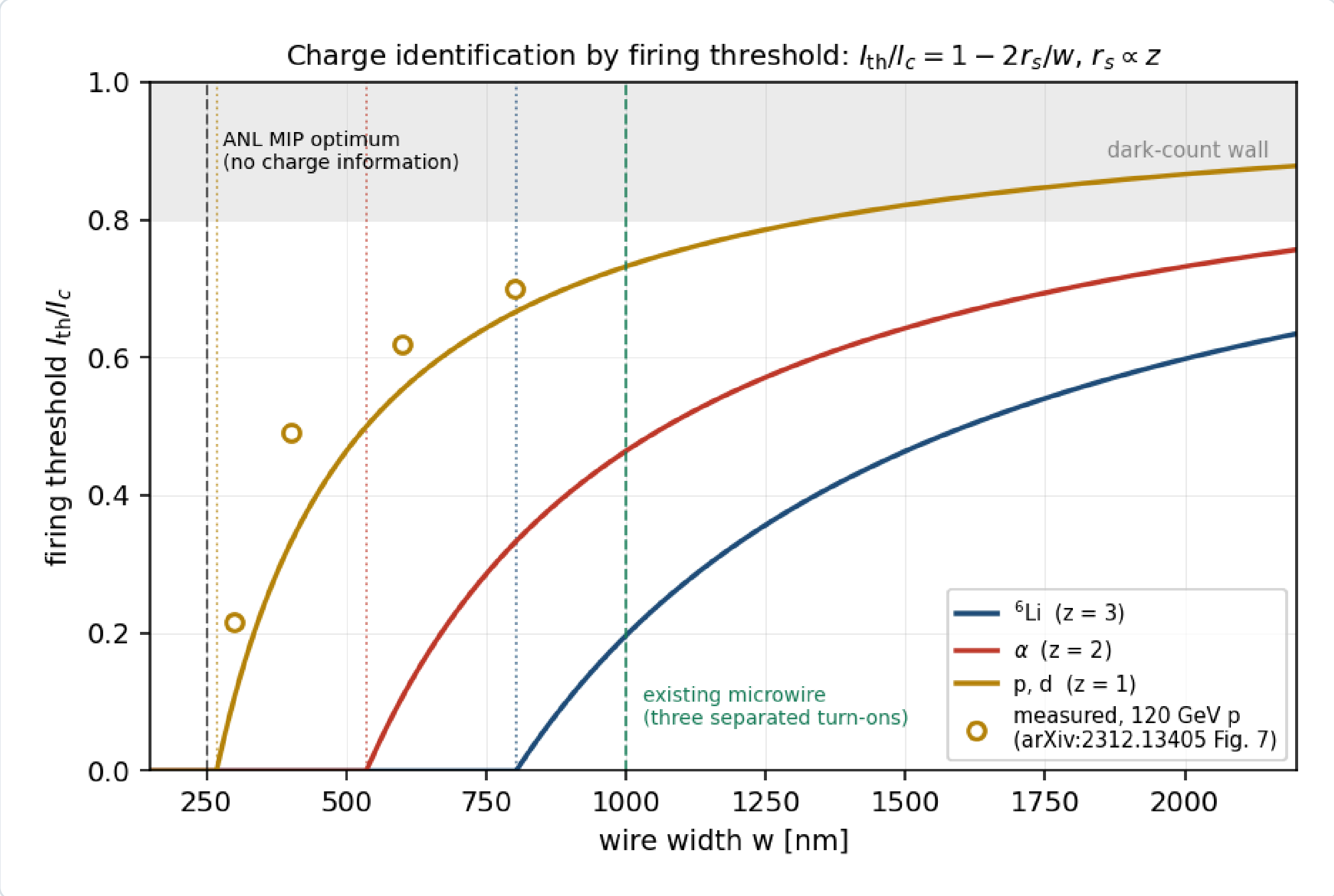

Figure 3 · Z by firing threshold. I_{th}/I_c against wire width for the three species, from $r_s \propto z$ anchored on Argonne's measured 134 nm for a 120 GeV proton. Open circles are their measured thresholds versus wire width, digitised from Fig. 7 of [4]. Dotted verticals mark $w = 2r_s$ for each species — below that width the threshold is zero, the wire fires at any bias, and it carries no charge information at all. The measured points sit above the z = 1 curve because inverting each of them for r_s gives 102–118 nm (mean 113) against the 134 nm Argonne quote as their extrapolated "hard" hot spot; every design number here is a *ratio* between species, so the 15% offset moves nothing — re-anchoring on 113 nm leaves 1 µm separating all three below the noise wall (`evgen/tests/test_nearbeam.py`). `nearbeam_sensor_budget.py` .

WIRE WIDTH	I_{TH}/I_C, P/D	A	^{6}LI	SEPARATED TURN-ONS
250 nm	0.00	0.00	0.00	**none** — ANL's MIP optimum
400 nm	0.33	0.00	0.00	**none**
800 nm	0.67	0.33	0.00	**α from d**
1000 nm	0.73	0.46	0.20	**all three** — the existing microwire
1500 nm	0.82	0.64	0.46	**all three**

At w = 1000 nm the three turn-ons sit at 0.20, 0.46 and 0.73 I_C. **Two bias points, at 0.33 and 0.60 I_C, tag Z by the firing pattern alone** — a Z = 3 plane that is blind to α and d, and a Z ≥ 2 plane that is blind to d — and both sit well below the 0.80 I_C at which Argonne measure the dark-count rate rising exponentially [4]. The scheme is not speculative in kind: bias-point charge-state discrimination is a granted patent and was demonstrated on singly- versus doubly-charged lysozyme by subtracting two bias points [9].

Three things make this the strong part of the case. (i) It is an interpolation, not an extrapolation. Argonne have measured a 120 GeV proton at r_s = 134 nm [4] and a 5.5 MeV ^{241}Am α, which their own √Q scaling puts at ≈ 1 µm [5]. A relativistic ^{6}Li lands at ≈ 400 nm — between their two measured points, on devices from the same cryostat programme. **(ii) The Z-ID operating point is also the blackbody-immune one.** A 300 K surface radiates ≈ 3×10^{18} photons cm^{-2} s^{-1} above the 0.04 eV SNSPD threshold, so a photon-mode device cannot look at a warm beam pipe at all. Biasing at 0.33 and 0.60 I_c puts the wire below its photon threshold and leaves only the hard-hot-spot particle mode — which is what the Z-ID scheme wants anyway. **(iii) The area answer and the Z-ID answer are the same device.** Argonne's efficiency optimum, w ≈ 250 nm, is precisely $2r_s$ for a proton: maximal MIP efficiency and *zero* charge information. Z-ID needs deliberately *wide* wires — which is the microwire (SMSPD) geometry that Caltech/JPL/Fermilab already fabricate and beam-test at 1 µm [7]. The device the charge measurement wants is the device the area problem also wants.

The honest limit. One threshold plane is a one-bit dE/dx measurement on a straggling distribution: an independent Landau-straggling estimate puts the α → ^{6}Li confusion at 20–25% per plane, improving only weakly with film thickness, so sub-percent mis-identification needs three to five independent planes. That is a limitation of any threshold dE/dx detector, not something peculiar to nanowires — and it is still the difference between a stack with charge information and a Roman Pot with none.

5 · What stands in the way

None of the following is a physics objection. All of them are engineering or programmatic, and all of them are load-bearing.

OBSTACLE	THE NUMBER	STATUS
Cold-stage power budget	A NbN device needs 2.8–3 K, a WSi microwire 0.8 K. SNSPD cold-stage loads are hundreds of µW to a few mW. A 4 K pulse tube delivers 1.5 W at 4.2 K and *zero* at its 2.8 K no-load floor, on a 190 kg water-cooled compressor drawing 10.7 kW.	hard
Beam-induced RF heating	A ferrite-shielded TOTEM Roman Pot dissipates ≈ 40 W at the LHC. Scaling the resistive-wall/broadband factor $Q^2M/\sigma_z^{3/2}$ to EIC parameters gives ≈ 16 / 8 / 0.8 W at highest-ECM / highest-L / 41 GeV. Even the 41 GeV case exceeds a pulse tube's entire 4.2 K lift.	hard
In-vacuum precedent	None found anywhere for a biased superconducting detector inside an accelerator's *beam* vacuum. The precedents — LHC CryoBLM silicon and diamond at 1.9 K, the CERN AD SQUID current comparator — sit outside it, and the AD beam is ~10^{7} antiprotons against the EIC's 8×10^{13} protons.	open
Vibration	Cryocooler cold-head displacements are specified at 7–9 µm; a purpose-built decoupling structure brings that to 10 nm but is heavy and stiff — the opposite of a pot that rides motorised rails to within 10σ of the beam every fill.	mitigable
Radiation hardness	Unmeasured. The FY22 proposal's central deliverable was *"an upper limit for the accumulated dose ... This currently unknown limit"*; four years on, no publication reports the LEAF irradiation, the JLab Hall C parasitic run or the SEU cross-section. The 2023 Quantum Sensors for HEP report says SNSPDs *"are expected to be radiation hard ... Although published studies in this area are lacking"*. The whole "radiation hard, therefore closer" argument is an anticipation — the proposal itself writes "(anticipated)".	unmeasured
Area	The largest SNSPD/SMSPD ever run in a GeV particle beam is 2 × 2 mm² with 8 channels [8]. A near-beam strip 3 mm deep over 20 mm of vertical extent, both sides of the beam, is 120 mm² — 30× the state of the art, and 1000× if the whole 9.3 mm gap is instrumented. Tiled as 30 × 30 µm² nanowire devices that is 133,000 channels; as 1 × 1 mm² microwire tiles it is 960.	funded R&D
Latching	Latching is a hard failure, not a dead time: the device locks resistive and stops detecting. Argonne's ^{241}Am α run is the reassurance — a ≈ 1 µm hot spot did *not* latch a 100–200 nm wire, and gave clean count-rate plateaus [5].	addressed
Rate	*Not* a blocker. The ePIC Preliminary Design Report puts pot rates at a few Hz per channel in normal running and *"30–50 kHz at ~10σ"* from beam halo, on STAR experience — four orders below the ~10^{8} counts/s a nanowire's fall time allows [6].	closed

Cryogenics also inverts the packaging argument. The nanowire's advantages — windowless operation in the beam vacuum, an active edge a few hundred nanometres from the substrate rim, no vacuum-window material — are real and are exactly what a near-beam sensor wants. But a device that cannot see a 300 K surface must be enclosed in a cold shield, and a cold shield is material in front of the sensor. The windowless argument and the blackbody constraint pull against each other, and the resolution — bias below the photon threshold, §4(ii) — is available but has never been demonstrated next to a beam.

6 · A baseline correction: the ePIC pot geometry has moved

Independent of nanowires, and more immediately actionable. The aperture this repository measured was taken in the September-2024 `epic-main` inside `jug_xl-nightly`. Reading the current `main` branch of `eic/epic` directly [3]:

	SEPTEMBER-2024 (WHAT `TOOLS/FULLSIM` MEASURED)	CURRENT `MAIN`
module	32 × 32 mm	16 × 16 mm
insertion	energy-*independent*: 3.2 cm outer, +0.7 cm central	per-energy 10σ offsets in `beamline_*.xml`
RF shield	1 mm aluminium on each module face, *active*	commented out — *"we don't know if we will even need it ... Oct. 2025"*
implied blind block at 18 × 275	32 mm (x) × 7 mm (y)	16 mm (x) × 2.7 mm (y)

The old geometry's 32 mm block gives 32 mm / 30.6 m = 1.046 mrad against the 1.03 mrad the scan measured — agreement to 1.5%, which is the check that the measurement and the file reading confirm each other. The current geometry's 16 mm block implies roughly half that, *below* the 0.727 mrad 10σ envelope; at 5 × 41 the per-energy insertion moves the other way, to a 29.6 mm inner edge.

> **What this means for the numbers in circulation.** The measured aperture, the sign-flipped $\langle\cos 2\beta\rangle = -0.77$, the 1.4×10^{-2} tagged fraction, and the conclusion that *"the coherent programme is a low-energy programme"* are all conditioned on a superseded geometry. They should be re-measured against the current release before being quoted further. The conversion from angle to millimetres also needs the per-optics R_{12}, which was measured for 18 × 275 alone (30.6 m); this note therefore works in angle throughout and quotes millimetres only for that configuration. **The curves of Figure 1 are unaffected** — they price every aperture, which is why they are plotted that way; what moves is only the marker labelled "you are here".

7 · What to ask, and of whom

#	QUESTION	OWNER	WHY IT DECIDES SOMETHING
D1	Turn-on bias current versus wire width, measured for the ^{241}Am α on the same wires as the 120 GeV proton — does the α curve sit where $r_s \propto z$ predicts?	ANL MEP	Validates the whole Z-ID case with data they already own. The α analysis is explicitly *"underway"* [5].
D2	Has the ~250 nm optimum ever been checked against anything but z = 1? Nobody has put a relativistic z > 1 nucleus in front of an SNSPD.	ANL MEP	If Z-ID is real, this is the first measurement of it — and FTBF or the SPS could do it with a light-ion beam.
D3	Re-measure the pot aperture on the current `epic-main`, per optics, with the RF shields as they now stand.	this repository (plans/09 B1)	Every aperture-conditional number depends on it (§6).
D4	Per-optics R_{12} and R_{34}, and the beam sizes σ_x, σ_y at the pot planes for a *lithium* fill.	ePIC FF WG / C-AD	The 10σ offsets exist only for proton optics, and are marked *"rough extrapolation"* at 5 × 41 [3]. This is plans/04 #20's original ask, still unanswered.
D5	What sensing area, at what wire width and channel count, by what year?	ANL MEP (Polakovic Early	Area is the blocker and the award is explicitly to *"expand their effective sensing areas"*. Armstrong's parallel award

		Career award, to 2029)	targets JLab ^{4}He, not the EIC pot — so an EIC-pot demonstrator is not currently anyone's funded near-term deliverable.
D6	A dose or fluence limit for NbN films, and the radiation environment at the ePIC pots to compare it against.	ANL MEP / ePIC FF WG	Neither number is published. Without both, "more radiation hard" cannot become "therefore N mm closer".
D7	Beam-coupling impedance and RF heating of the pot assembly.	C-AD	No quantitative EIC pot impedance study is public; the geometry comment says the shield's necessity is unknown for exactly this reason [3]. It sets the cryogenic load (§5).

8 · Conclusion

The case for superconducting nanowires at the ePIC far-forward is real but it is not the case that is usually made for it. The aperture argument — radiation-hard sensors move closer, acceptance follows — has the right destination and the wrong mechanism: the 10σ rule protects the machine, not the sensor, and a cryostat is a larger object than an AC-LGAD stave, not a smaller one. What is worth a great deal is *any* closer approach, by whatever means: Figure 1 prices it, and at 10 × 100 it is the difference between a measurement and a non-measurement.

The argument that does hold up is charge identification. Open question #19 has no answer at IP6, the physics of the firing threshold supplies one, the wire width it needs is the width the area problem independently pushes toward, the operating point it needs is the one that makes the device immune to a warm beam pipe, and the calibration for it brackets our signal between two measurements Argonne have already made. That is a specific, testable, near-term ask — and it is worth putting to the group ahead of anything about apertures.

References

[1] R. Abdul Khalek et al., *Science Requirements and Detector Concepts for the Electron-Ion Collider: EIC Yellow Report*, Nucl. Phys. A 1026 (2022) 122447, arXiv:2103.05419, §14.5.3 (`refs/2103.05419_part4.pdf`): the four EIC SNSPD applications, of which the first is the far-forward Roman Pot; edgeless operation "within a few 100 nm of the substrate edge"; membranes "as thin as few 10 μm"; the mm² pixel-array goal.

[2] W. Armstrong (PI) et al., *Superconducting Nanowire Detectors for the EIC*, FY22 EIC-Related Generic Detector R&D proposal #18, §3.3 Concept 1 — reviewed and partially funded (Report of the 2022 EIC-related Generic Detector R&D Review Committee, June 2023). The 10σ rule-of-thumb passage quoted in §3.

[3] `eic/epic` , branch `main` : `compact/far_forward/roman_pots_eRD24_design.xml` and `compact/fields/beamline_{18x275,10x100,5x41}.xml` , read directly (2026-08-26). Module 1.6 × 1.6 cm; per-energy offsets with the comment "These are the ten-sigma cuts for the Roman pots, translated to the physical layout we currently have. They are not perfectly ten-sigma for reasons of physical geometry."; RF shield module components commented out, "Oct. 2025".

[4] S. Lee, T. Polakovic, W. Armstrong, A. Dibos, T. Draher, N. Pastika, Z.-E. Meziani, V. Novosad, *Beam Tests of SNSPDs with 120 GeV Protons*, Nucl. Instrum. Meth. A 1069 (2024) 169956, arXiv:2312.13405 (`refs/2312.13405.pdf`). 12 nm NbN, 30 × 30 μm², wire widths 300–800 nm at 2.82 K; Eqs. 1–2; r_s = 134 nm; ~250 nm optimum; background rising exponentially above I_b/I_c = 0.8.

[5] S. Lee, W. Armstrong, J. DiPreta, C. Dulya, V. Novosad, T. Polakovic, *Optimization of Cryogenic Detector Test Station by Rejecting Electromagnetic Interference*, Nucl. Instrum. Meth. A 1093 (2027) 171953, arXiv:2601.03158 (`refs/2601.03158.pdf`). The 5.5 MeV ^{241}Am α at ≈ 1 μm hot spot by the same √Q scaling; clean plateaus with no latching; "a detailed study of α detection is underway".

[6] ePIC Collaboration, *Preliminary Design Report*, 29 September 2024, the Roman Pots and off-momentum detectors section: "Rates during normal operations, with expected vacuum of 10^{-9} mbar, are a few Hz/channel. However, the beam halo could potentially provide rates of 30–50 kHz at ~10σ from experience of Roman pots at STAR."

[7] C. Wang et al. (Caltech / JPL / Fermilab), *Towards High-Efficiency Particle Detection Using Superconducting Microwire Arrays*, arXiv:2510.11725 (`refs/2510.11725.pdf`). 8-channel 1 × 1 mm² WSi, 4.7 nm film, 1 μm wires on a 3 μm gap (25% fill), 0.8 K; 75% fill-factor-normalised efficiency and 130 ± 17 ps at CERN SPS H6.

[8] C. Peña, C. Wang, S. Xie et al., *Characterization of a Superconducting Microwire Single-Photon Detector Array for Charged-Particle Detection*, JINST 20 (2025) P03001, arXiv:2410.00251 (`refs/2410.00251.pdf`). 2 × 2 mm², 8 channels, 30 ± 1 μm spatial resolution; amplitude distributions indistinguishable across species; sensor edge sharper than the 10 μm telescope.

[9] R. Cristiano, M. Ejrnaes, A. Casaburi, N. Zen, M. Ohkubo, *Superconducting nano-strip particle detectors*, Supercond. Sci. Technol. 28 (2015) 124004: the hot-spot threshold in the ion literature; the charge-state discrimination demonstrated on lysozyme by bias-point subtraction (Suzuki et al., Rapid Commun. Mass Spectrom. 24 (2010) 3290; US Patent 8,872,109 B2, Ohkubo & Suzuki, AIST). Also J. J. Renema et al., arXiv:1301.3337, for the photon-side form of the same relation.

[10] T. Polakovic, W. R. Armstrong, V. Yefremenko, J. E. Pearson, K. Hafidi, G. Karapetrov, Z.-E. Meziani, V. Novosad, *Superconducting nanowires as high-rate photon detectors in strong magnetic fields*, Nucl. Instrum. Meth. A 959 (2020) 163543, arXiv:1907.13059 (`refs/1907.13059.pdf`). Saturated efficiency to 5 T *parallel* to the device plane but only ≈ 0.5 T perpendicular; 10^7 counts/s measured, ~10^8 /s from τ_F = 11.78 ns.

Appendix A · Revision history

DATE	REVISION
2026-08-26	First version. The aperture scan and the coherent chain at both apertures (Figures 1–2, `nearbeam_aperture_scan.py` , `nearbeam_reach_gain.py` , and `--near-beam-mrad` added to `money_cos2phi_coherent_reco.py`); the hot-spot firing-threshold model of charge identification (Figure 3, `nearbeam_sensor_budget.py`); the obstacle table; the ePIC geometry correction of §6, read from the current `main` branch; the question list of §7.

All acceptances use B = 50 GeV⁻² for the coherent |t| slope and the high-acceptance optics (σ_θ = 72.7 μrad, 10σ = 0.727 mrad). The measured silicon aperture is from `tools/fullsim` in the September-2024 `epic-main` — see §6. Energy deposits are Bethe means in a 12 nm NbN film; the Landau most-probable value is deliberately not quoted, because ξ/I ≈ 0.002 in this film puts it far outside the ξ ≫ I regime the formalism requires. Hot-spot radii are anchored on Argonne's measured 134 nm and scaled as z. Built by `reports/build_report.py` ; figures from the `evgen/scripts` named in the text; display mathematics typeset at build time.